OS & SP

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SKILL – 12

1.Skill12.c

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GNU nano 8.7.1 skill12.c *

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main(void)
{
    int pipefd[2];

    /* Create pipe */
    if (pipe(pipefd) == -1) {
        perror("pipe");
        return 1;
    }

    printf("Pipe created successfully.\n");
    fflush(stdout);

    /* Create child process */
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        close(pipefd[0]);
        close(pipefd[1]);
        return 1;
    }

    /* Child process */
    if (pid == 0) {
        close(pipefd[1]);

        /* Redirect pipe input to stdin */
        if (dup2(pipefd[0], STDIN_FILENO) == -1) {
            perror("dup2");
            close(pipefd[0]);
            _exit(1);
        }

        close(pipefd[0]);

        printf("Child: Reading data from pipe...\n");
        fflush(stdout);

        char buffer[100];
        ssize_t n = read(STDIN_FILENO, buffer,
                         sizeof(buffer) - 1);

        if (n > 0) {
            buffer[n] = '\0';
            printf("Child received: %s\n", buffer);
        } else if (n == -1) {
            perror("read");
            _exit(1);
        }
        _exit(0);
    }
    /* Parent process */
    close(pipefd[0]);
```

```c
    /* Parent process */
    close(pipefd[0]);
    char message[] = "Hello from parent process!";
    printf("Parent: Data written to pipe.\n");
    if (write(pipefd[1], message, sizeof(message)) == -1)
        perror("write");
    /* Close write end */
    close(pipefd[1]);
    /* Wait for child */
    waitpid(pid, NULL, 0);
    printf("Parent: Child process completed.\n");
    return 0;
}
```

OUTPUT :

```
lohitha27@LAPTOP-A9F545FG:~/SKILL$ nano skill12.c
lohitha27@LAPTOP-A9F545FG:~/SKILL$ gcc -Wall -Wextra -g skill12.c -o skill12
lohitha27@LAPTOP-A9F545FG:~/SKILL$ ./skill12
Pipe created successfully.
Parent: Data written to pipe.
Child: Reading data from pipe...
Child received: Hello from parent process!
Parent: Child process completed.
lohitha27@LAPTOP-A9F545FG:~/SKILL$
```